

Reviewer Pack — Minimal Local Induction Ring Rank and SAT Clause Subset Comp...

Entry b2e575bf6928 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Induction Ring Rank and SAT Clause Subset Complexity

Entry ID: b2e575bf6928 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 20:57:23 UTC

1. Conjecture

Field A (mathematical branch): Algebra (Local Induction Rings) **Field B** (complexity object): Boolean Satisfiability (SAT Clause Subset Complexity)

Statement:

For every Boolean satisfiability instance with n clauses, the minimal local induction ring rank of its induced poset is linearly correlated with the complexity of computing a clause subset of size at most $\log(n)$, such that $MLI(n) = \Theta(\log(n) * c(n))$ for some constant $c(n)$.

Rationale (proposer's reasoning):

Local Induction Rings (LIRs) provide a framework for studying algebraic structures in relation to their computational properties. This conjecture posits that the complexity of computing a clause subset is tightly linked to the algebraic structure encoded by the LIR, suggesting potential insights into the inherent hardness of SAT.

Taxonomy category: LOCAL_INDUCTION_RINGS (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 8acc2645045b31f0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all n clauses ($n \leq 40$), the correlation coefficient between the minimal local induction ring rank and the computational complexity of finding a clause subset of size at most $\log(n)$ exceeds 0.9, with $p\text{-value} < 0.05$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Local Induction Ring Rank AND Boolean Satisfiability - Induced Poset Complexity in Algebra SAT Clause Subset - Local Induction Rings and SAT Clause Subset Size Relationship

Top relevant hits considered: - [<http://arxiv.org/abs/1607.05594v2>] Poincaré series of compressed local Artinian rings with odd top socle degree - [<http://arxiv.org/abs/astro-ph/0405128v1>] Probing the Evolution of the Galaxy Interaction/Merger Rate Using Collisional Ring Galaxies - [<http://arxiv.org/abs/1911.04516v1>] Boolean lattices in finite alternating and symmetric groups - [<http://arxiv.org/abs/1908.01624v1>] Learned Clause Minimization in Parallel SAT Solvers - [<http://arxiv.org/abs/2207.13577v2>] Scalable Proof Producing Multi-Threaded SAT Solving with Gimsatul through Sharing instead of Copying Clauses - [<http://arxiv.org/abs/math/9712256v2>] Hopf Algebras and Edge-Labeled Posets - [<http://arxiv.org/abs/1505.03340v2>] HordeSat: A Massively Parallel Portfolio SAT Solver

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_instance(n):
        clauses = []
        for _ in range(n):
            clause = [random.choice([True, False]) for _ in range(2)]
            clauses.append(clause)
        return clauses

    def min_local_induction_ring_rank(clauses):
        # Placeholder implementation
        return random.randint(1, n)

    def complexity_of_clause_subset(clauses, subset_size):
        # Placeholder implementation
        return len(subset_size) * 0.5

    n_values = [5, 10, 15, 20, 30, 40]
    results = []
```

```

for n in n_values:
    instance = generate_instance(n)
    mli_n = min_local_induction_ring_rank(instance)
    subset_size = int(math.log(n))
    complexity = complexity_of_clause_subset(instance, subset_size)

    if mli_n == 0 or complexity == 0:
        continue

    results.append({
        "metric_name": "MLI(n) * complexity",
        "metric_value": mli_n * complexity,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    })

return results[0] if results else {
    "metric_name": "MLI(n) * complexity",
    "metric_value": None,
    "instances_tested": 0,
    "n_max": 0,
    "conjecture_holds": False,
    "counterexample": "mapping_undefined"
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17,

    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    if all(result["conjecture_holds"] for result in results):
        mean_value = sum(result["metric_value"] for result in results) / len(results)
        support_fraction = 1.0
        print(f"RESULT: SUPPORTED mean={mean_value} std=0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next((result["seed"] for result in results if not result["conjecture_holds"]))
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5f8b82af.py", line 72, in <module>
    trial_result = run_trial(seed)

```

```

^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5f8b82af.py", line 43, in run_trial
    complexity = complexity_of_clause_subset(instance, subset_size)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5f8b82af.py", line 34, in complexity_of
    return len(subset_size) * 0.5
    ^^^^^^^^^^^^^^^^^
TypeError: object of type 'int' has no len()

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means that the pre-registered support condition could not be unambiguously met. | next: Investigate and fix the error in the test code to ensure it can run to completion and produce results for further analysis.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	19616
2	propose	ollama_remote	glm4:latest	0	0	13051
3	preregistration	ollama_remote	glm4:latest	0	0	9082
4	novelty	ollama_remote	glm4:latest	0	0	10737
5	novelty	ollama_remote	glm4:latest	0	0	9845
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	46870
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11376
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9392
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7580
10	judge	ollama_remote	glm4:latest	0	0	13187

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 150737 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/b2e575bf6928.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/b2e575bf6928.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/b2e575bf6928.tar.gz` (if generated)